

# Potato Disease Management

Covers: Late Blight, Early Blight, Healthy

Compiled from public university cooperative extension publications for testing InsightAI-RAG's retrieval and grounding quality against real agricultural guidance. This is not a substitute for consulting a local agronomist or extension office before applying any treatment. Always follow current product labels; pesticide registrations change and vary by region.

## Late Blight — *Phytophthora infestans*

A highly destructive oomycete (water mold) disease. Leaf symptoms are circular-to-irregular brown lesions surrounded by pale/chlorotic tissue; enlarged, water-soaked lesions show white-to-gray fuzzy sporulation, especially visible on the leaf underside. Tubers develop sunken, firm brown lesions extending several centimeters into the flesh. Disease spreads via wind-blown or splashed sporangia; under warm, humid conditions (90-100% humidity, 54-75F) infection can occur in as little as 2 hours, with new sporulation in 4-5 days, driving rapid, large-scale epidemics.

**Management:** Plant only healthy, certified seed tubers. Destroy cull tubers and volunteer plants from previous seasons, and destroy unmanaged infected plants promptly — the pathogen overwinters in infected tubers, not free in the soil. Space rows to increase airflow and reduce leaf wetness. Plant resistant cultivars where available. Apply fungicides preventively; consult a current regional guide for specific product recommendations, since efficacy and registration change.

Source: University of Wisconsin-Extension, 'Potato Late Blight: Identification and Management' (A4052-02)

## Early Blight — *Alternaria solani*

The most characteristic symptom is a 'bull's-eye' pattern of concentric-ring lesions on leaves and stems, starting small and enlarging until spots merge. Early blight typically infects the oldest, lowest leaves first and progresses upward as those leaves droop and dry out; it is generally a disease of stressed or senescing plants rather than vigorous young growth.

**Management:** Maintain plant vigor through balanced fertilization — stressed and nutrient-deficient plants are far more susceptible. Rotate away from potato and other solanaceous crops for at least 2 years. Avoid overhead irrigation and improve airflow to reduce leaf wetness duration. Remove and destroy infected debris at season's end. Fungicides containing chlorothalonil or mancozeb are commonly used where chemical control is warranted; follow current label directions.

Source: University of Minnesota Extension, 'Early blight in tomato and potato'; UC Statewide IPM Program, 'Early Blight / Potato' Pest Management Guidelines

## Healthy

No signs of disease detected. Continue routine scouting, particularly after periods of wet or humid weather when both early and late blight develop fastest.

Source: General extension guidance